A Data Scientist's greatest asset is curiosity, not just the technical skills. In an age where our technology is increasing production efficiency and the human input transforms to effortless or minimal. Our tools are doing the heavy lifting but the thinking, the creativity, the direction and the curiosity still comes from us. Tools will always evolve to make what we do easier, something I had to come to terms with as an aspiring data scientist. As I felt as if A.I. would have made my aspired career field obsolete, while it may affect the approach and the how things are done for many, autonomy still belongs to the humans.

My project was born out of my interest in the "behind-the-scenes" of films; the "what happens in the decision room" for a film. The decision to create a film I believe has 2 aspects to it which are the Business and the Creative. We can argue on whether the chicken or the egg was first but one factor is for sure that both the business and the creative are reliant on each other for if there is no investment to create a film there is no creative process to get it done and if there is no creative process then the business has no business goals(the number one goal is profit) that will be actualized. How I arrived at the curious question that started this project was I came across a dataset of movies on Kaggle sourced from TMDB and IMDB contributed by the Kaggle user asaniczka. After reviewing the data the questions started flowing but I wanted to approach the dataset with order and as if it were a real commissioned data project .

This order is the Data Science Methodology by John Rollins (shout out to IBM and Coursera for the Data Science Methodology course). It entails structuring a data project in the following data science life cycle order: Business Understanding, Analytic Approach, Data Requirements, Data Collection, Data Understanding, Data Preparation, Modelling, Evaluation, Deployment and Feedback. Let's get into it.

Cinema Gold (Predictive Modeling of Film Success)

Business Understanding (Define the problem, goals and desired outcome)

I approached the problem with a real world scenario. A movie studio wants to make a movie and wants to predict its best chances of being profitable.Create a model that can predict a film's profitability before it is released with an accuracy of 90%.

Naturally I asked the following questions: Can we predict a film's success before its release and what makes a film successful ?

Analytic Approach (Determine the technical method to achieve the goal)

Understanding the problem and the desired outcome informs the analytic solution to the business problem.

A supervised learning model is what I choose for this problem. How a supervised learning algorithm works is by observing data and being told what it is or what it should do with it. There are many types of supervised learning algorithms to choose from but I will be exploring just 2. Classification and Regression algorithms are viable options to approach the problem at hand.

The Classification algorithm will determine whether the film will become a success, yes or no while the Regression approach aims to predict the ROI an unreleased film can produce. I am using both to see which of the 2 approaches would yield an accuracy of 90% as desired in our outcome.

Data Requirements (Determines the data type, sample size, source and storage)

I need the right type of data to create an accurate model. I am using Google Colab as my base of operation for my project notebook and so I am aiming for the most portable/lightweight and flexible access method to the data I need.

- Source: IMDB & TMDB joined Movie Dataset via Kaggle API
- Storage: CSV File
- Data type: Text, Numbers and Date object
- Data Points: Movie Title, Genre, Release Year, Budget, Revenue, Profit, Production Studio

Data Collection (Create the data pipeline or gathering method)

I used the Kaggle Hub API as the foundation of my data pipeline to access the data I needed for this project and store the raw data into a csv file.

Please note: The movie data from The Movie Database (TMDb) has been sourced from Kaggle, contributed by asaniczka. TMDb provides a comprehensive API with data on movies, TV shows, and more, which is freely available for non-commercial use.

The IMDb movie data has been sourced from the IMDb Non-Commercial Datasets available at IMDb Non-Commercial Datasets. IMDb provides subsets of movie data for personal and non-commercial use, and this dataset is used with permission from IMDb under their Non-Commercial License. Please refer to IMDb's terms and conditions for more details on the usage restrictions.

Data Understanding (Explore quality, relevance and usability/EDA)

Now I dive into the details to understand what we have to work with for the predictive model. In the EDA (exploratory data analysis) process on the data I found that I have:

- 436,167 rows and 29 columns of data.
- Columns with 1,999,310 empty cells
- 2 Columns I would need to convert into categorical status
- I Identified all required data points except for Profit

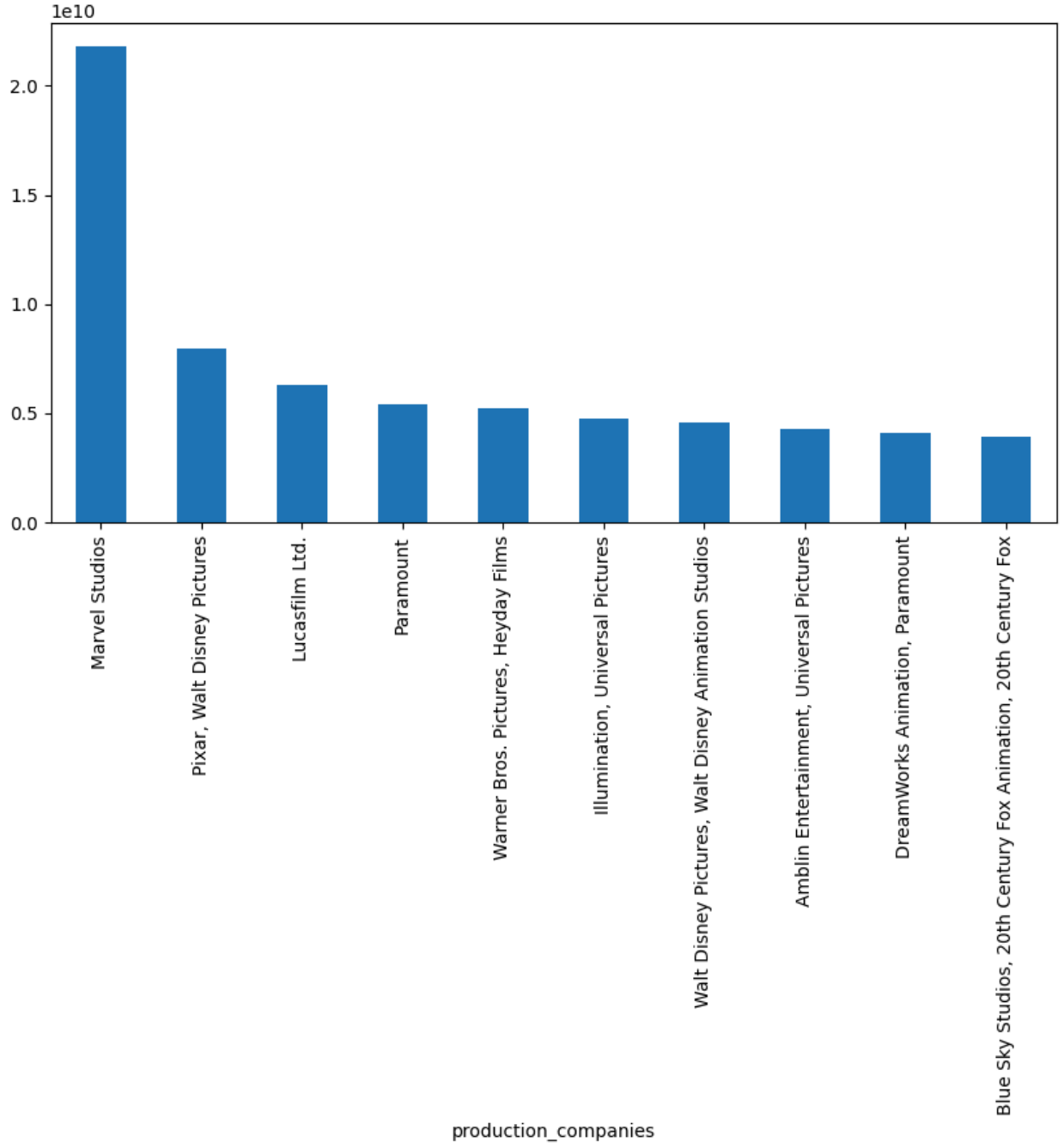
My next move was to sort the dataset by revenue just to see which movies off the bat had the highest revenue and all my favorites were in the top 10:

- Avatar
- Avengers: Endgame
- Avatar: The Way of Water
- Titanic
- Star Wars: The Force Awakens
- Avengers: Infinity War
- Spider-Man: No Way Home
- Jurassic World
- The Lion King
- The Avengers

I will leave you to guess which films are my favorites. If we know the studios and the directors behind these films we can already see who is dominating the space. Directors such as James Cameron, The Russo Brothers and Production Studios such as Marvel Studios and 20th Century Studios are repeat offenders. Keep this in mind for later tho.

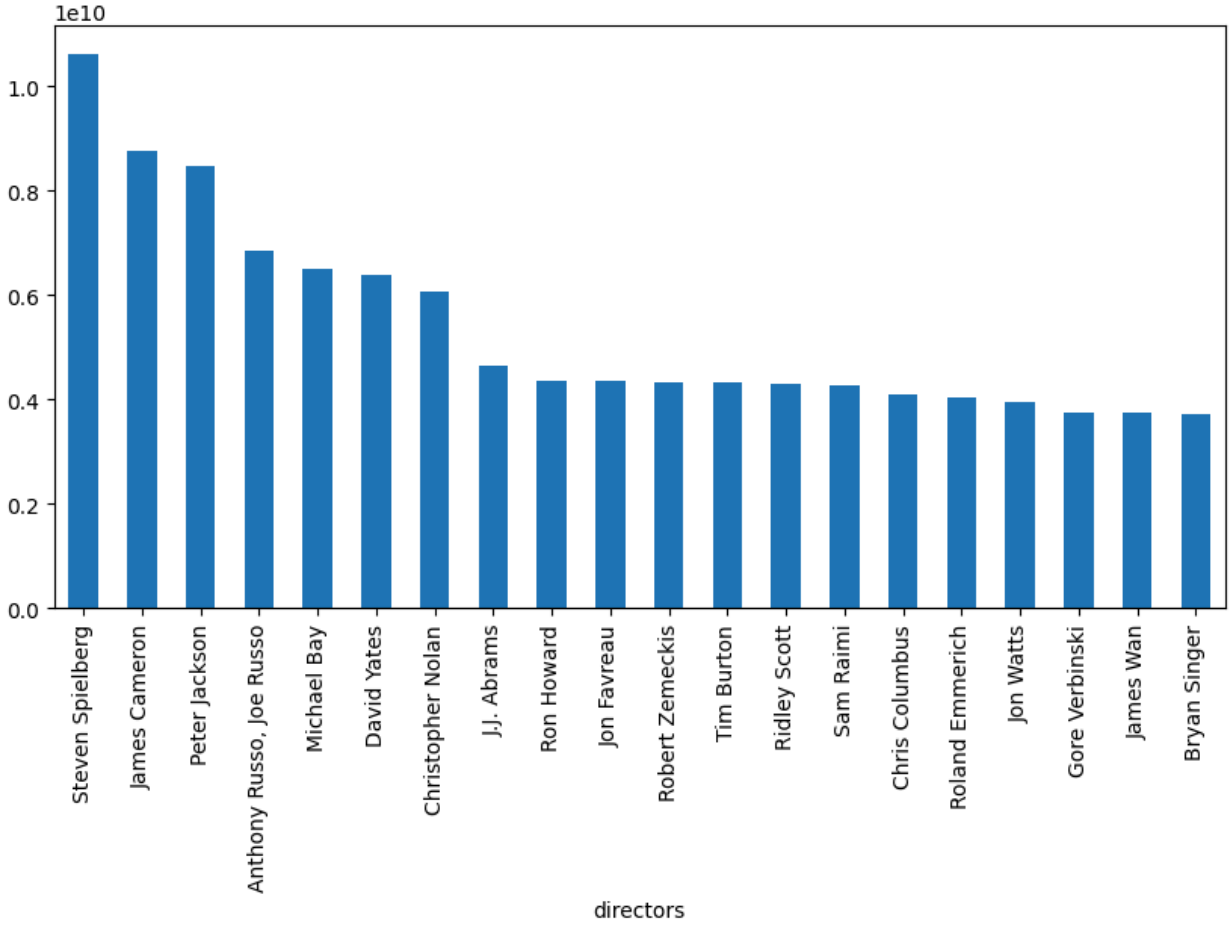
I continued to group the revenue together by different attributes as seen below:

Top 10 studios with the highest revenue.



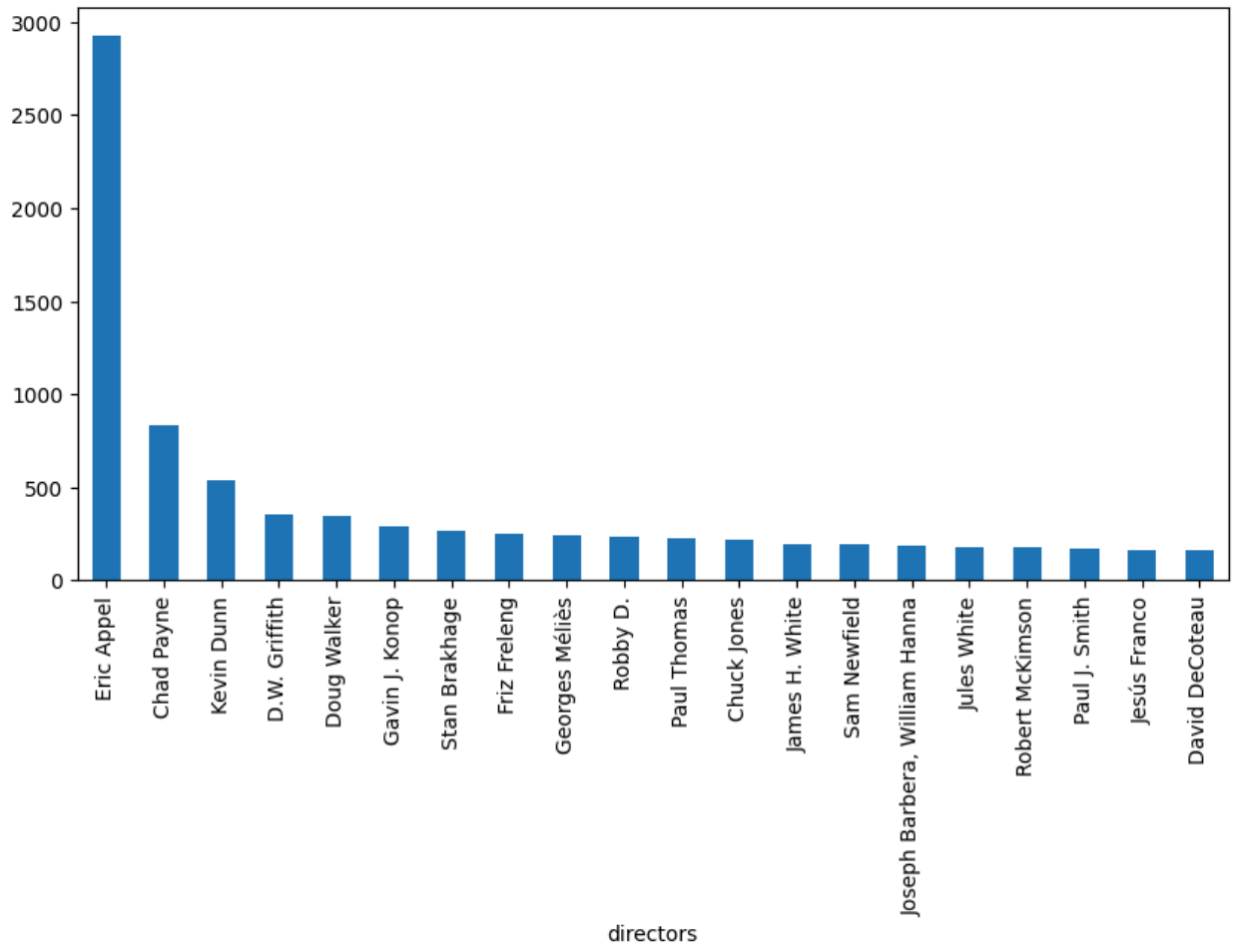
Marvel Studios is eating everything up easily in Jamaica we say “by donkey lengths” context clues will help you out if you're not familiar with the phrase.

Top 20 Directors with the highest grossing films



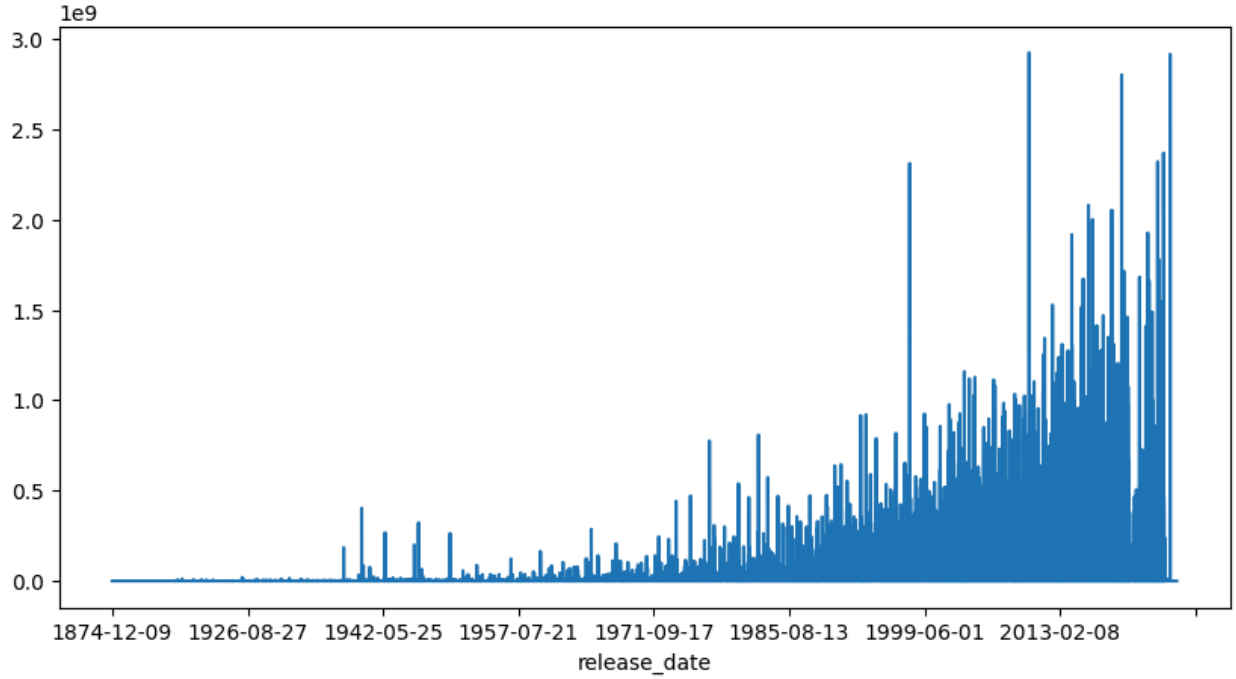
Steven Spielberg, James Cameron and Peter Jackson are the top 3 directors with the highest gross of their combined films.

Directors that made the most films



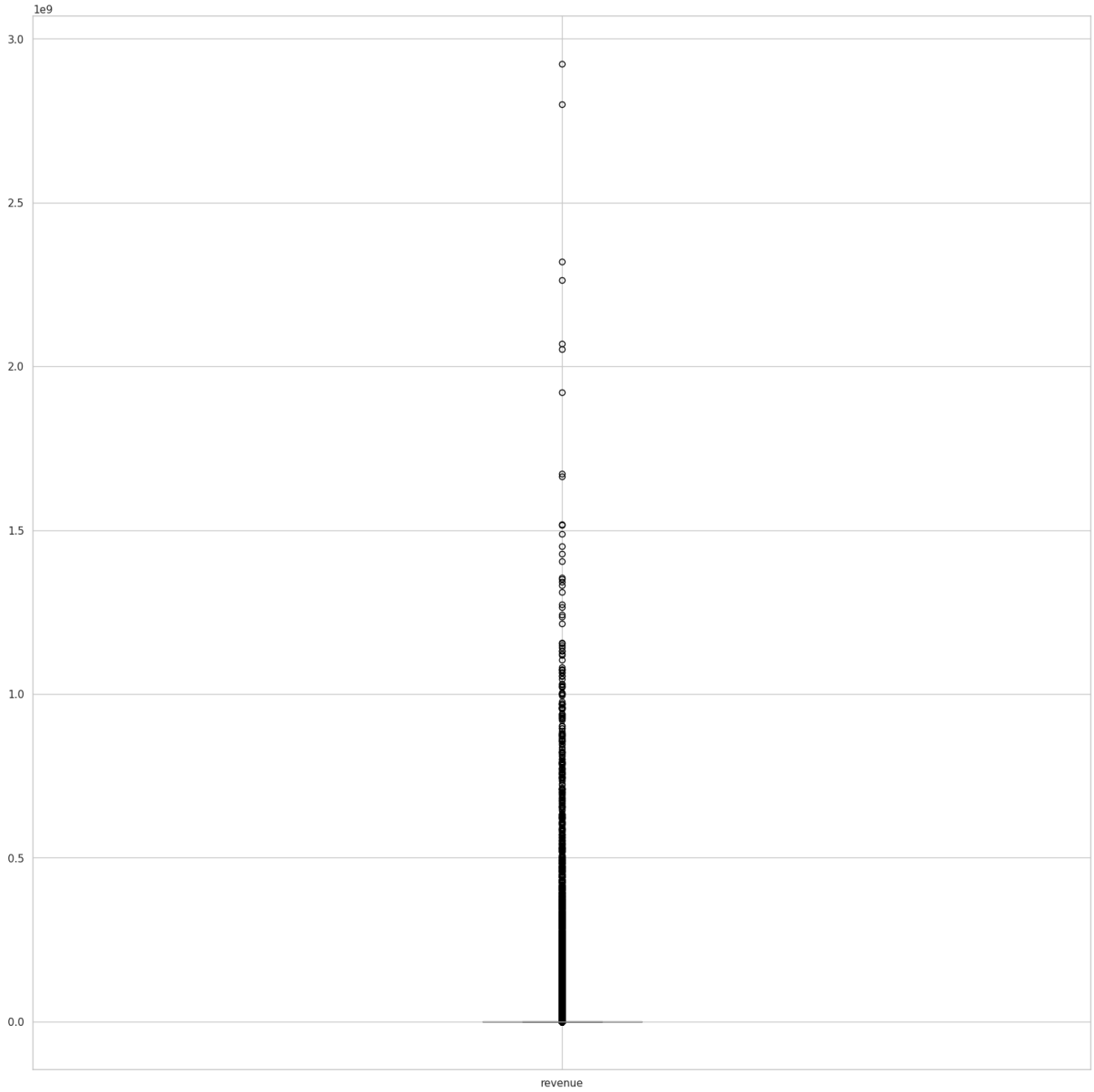
This is when it clicked for me, I don't recognize any of these names nor are any of these names on the list of directors that have made the most money with their films. So I then looked at the revenue by film release date.

Revenue by film release date



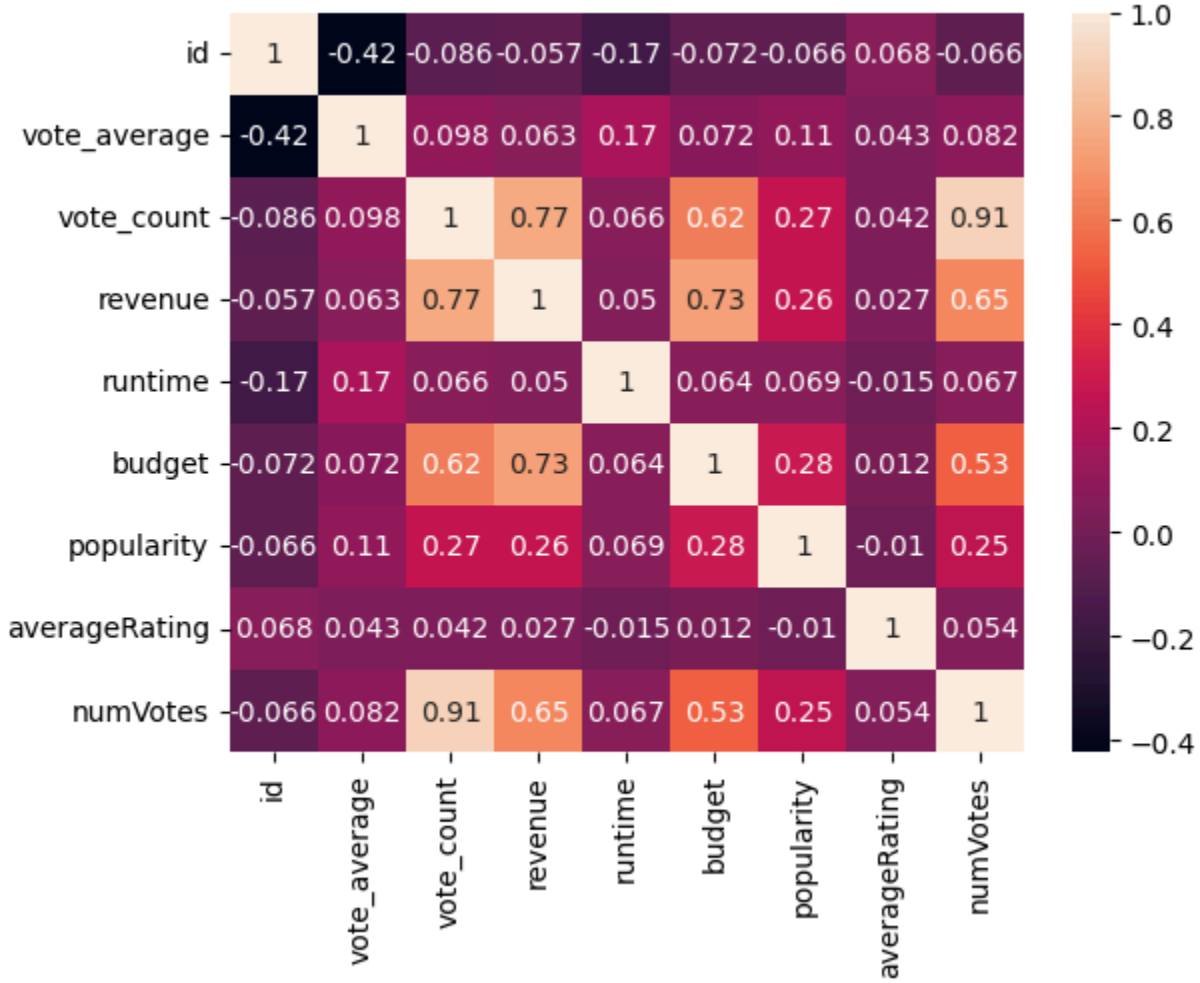
Or more like I tried to look at the revenue by release date because I came across another revelation, the release date datapoint isn't in date format and inflation isn't accounted for when comparing the revenue generated by films and directors as this dataset has movies from 1913 - 2024. So these are important things to note for the data preparation stage of the project.

I wanted to see if there were gaps in the revenue so I did a boxplot:



This is showing me that there are a few heavy hitters in gross revenue compared to the majority.

My next move was to explore the relationships the other columns had to ‘revenue’ in the dataset. We achieve by checking the correlation value and that is visualized with a heatmap. The lighter the color is, the stronger the relationship and darker the color is the weaker the relationship.



Revenue has a significant positive relationship, they are: ‘numVotes”, ‘popularity’, ‘budget” and ‘vote_count”.

Data Preparation (Conduct ETL, Feature Engineering Normalization)

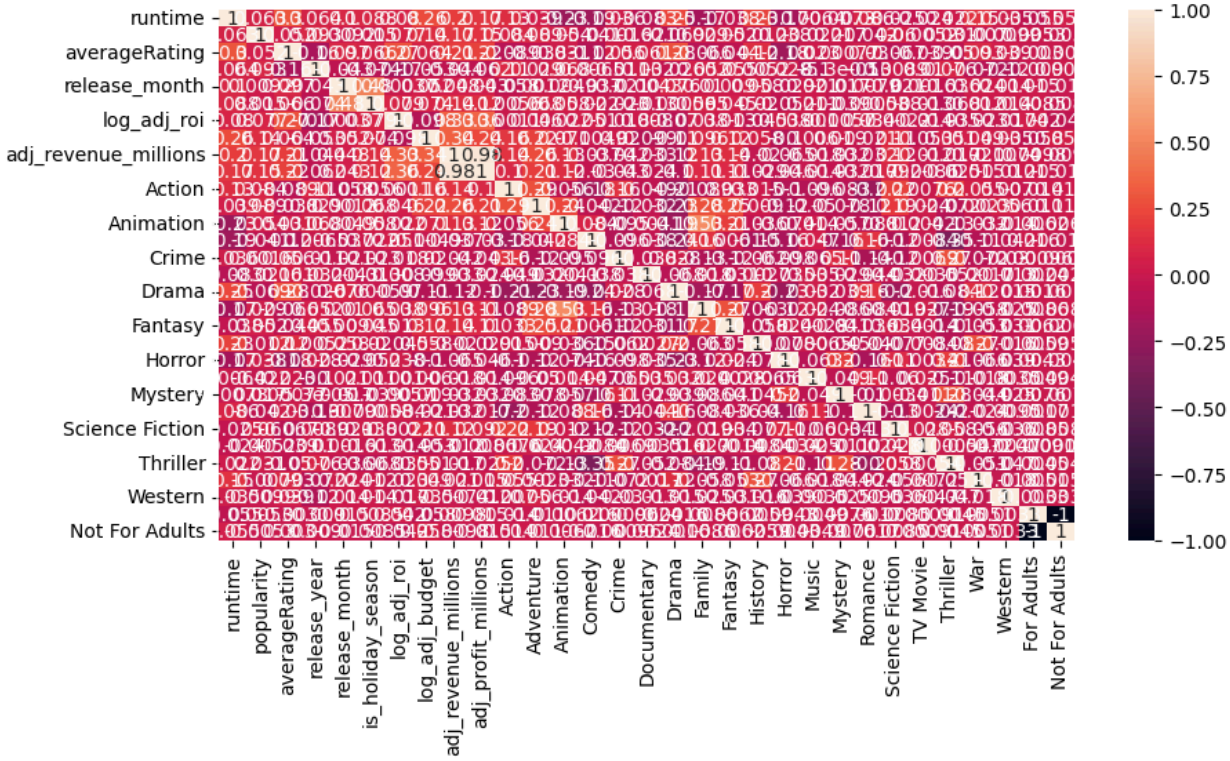
With all I have uncovered in the data understanding stage I had my work cut out for me. I had a lot of mess to sweep up and organize to do.

I handled the data in this following order:

- Removed rows with missing data
- Removed columns that had no positive correlation to revenue
- Feature Engineered release year and release month columns from release date
- Created a a holiday season column by gathering June, July, November and December
- Calculated Profit, ROI
- Used AI to generate a Customer Pricing Index table so I can adjust our revenue, profit and budget values for inflation. This would make the money truly comparable for films in different periods when the dollar value was different.
- Normalized the ROI and Budget values to reduce outlier effects on the model.
- I then converted the genre into categorical data represented by 0s and 1s in a process called One Hot encoding, the same was also done for the made for adult column

After I completed this stage I was left with 9327 rows and 57 columns of data from 436,167 rows and 29 columns. I saved a new csv file of clean data for the predictive model to be trained and tested on.

Afterwards I wanted to see what the correlation looked like at that point.



It was too much for me to make sense of at this stage. This is where I express gratitude for these tools we use because I gave it a try but left it up to the model to tell me which of these features was the most important for making the prediction.

Modelling (The algorithm formation from learned data patterns)

Here is where we get to the meat of the matter, the reason we gathered, creating the solution to our problem, can we predict a film's success before its release ? What makes a film successful ? I will try my best not to get too technical with my words at this stage and beyond. Our chosen supervised learning methods are Classification and Regression algorithms. The goal is to predict a film’s success before its release 9 times out of 10.

Each model was trained using 80% of the dataset and tested with the remaining 20%. The features used to train the models are Budget', Runtime', Average Rating, Release Month, Release year, Seasonality (June, July, November and December), Movie Genre, For adult, not for adult data points. The target for both models are different but I will specify when I get there.

Regression Model

The regression algorithm used was a RandomForestRegressor and its target feature was to predict ROI.

Classification Model

The classification algorithm used was a RandomForestClassifier and I had to engineer its target feature. Its target was ‘is_hit” defined by the ROI being 100% or more which tells the model if the ROI is less than 100% the film did not double its investment in revenue and therefore considered a flop but if it achieved 100% or more the film can be considered a hit or success for the studio.

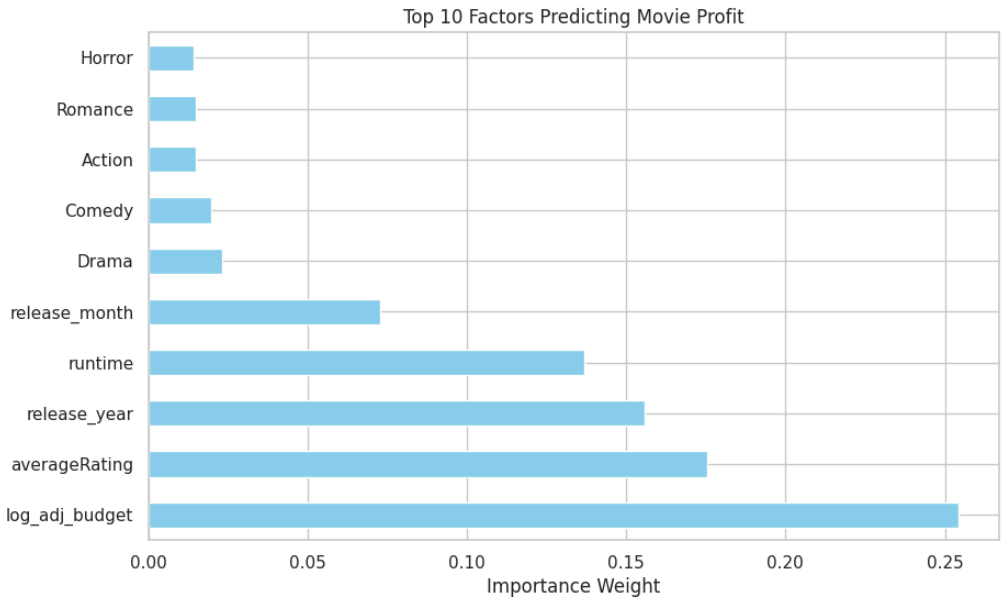
As it pertains to the outcome for each of these models we will dive into that in the next stage.

Evaluation (Assess statistical significance, accuracy and success criteria)

This stage was exciting for me and long awaited because most of my time was spent on the Data Preparation stage and now the model has been trained its time to get the report on well the models performed with that test data after its training.

Regression Model Evaluation

Evaluation Metrics:
- Average Prediction Error: $1.10 Million
- Accuracy Score (R2): 27%

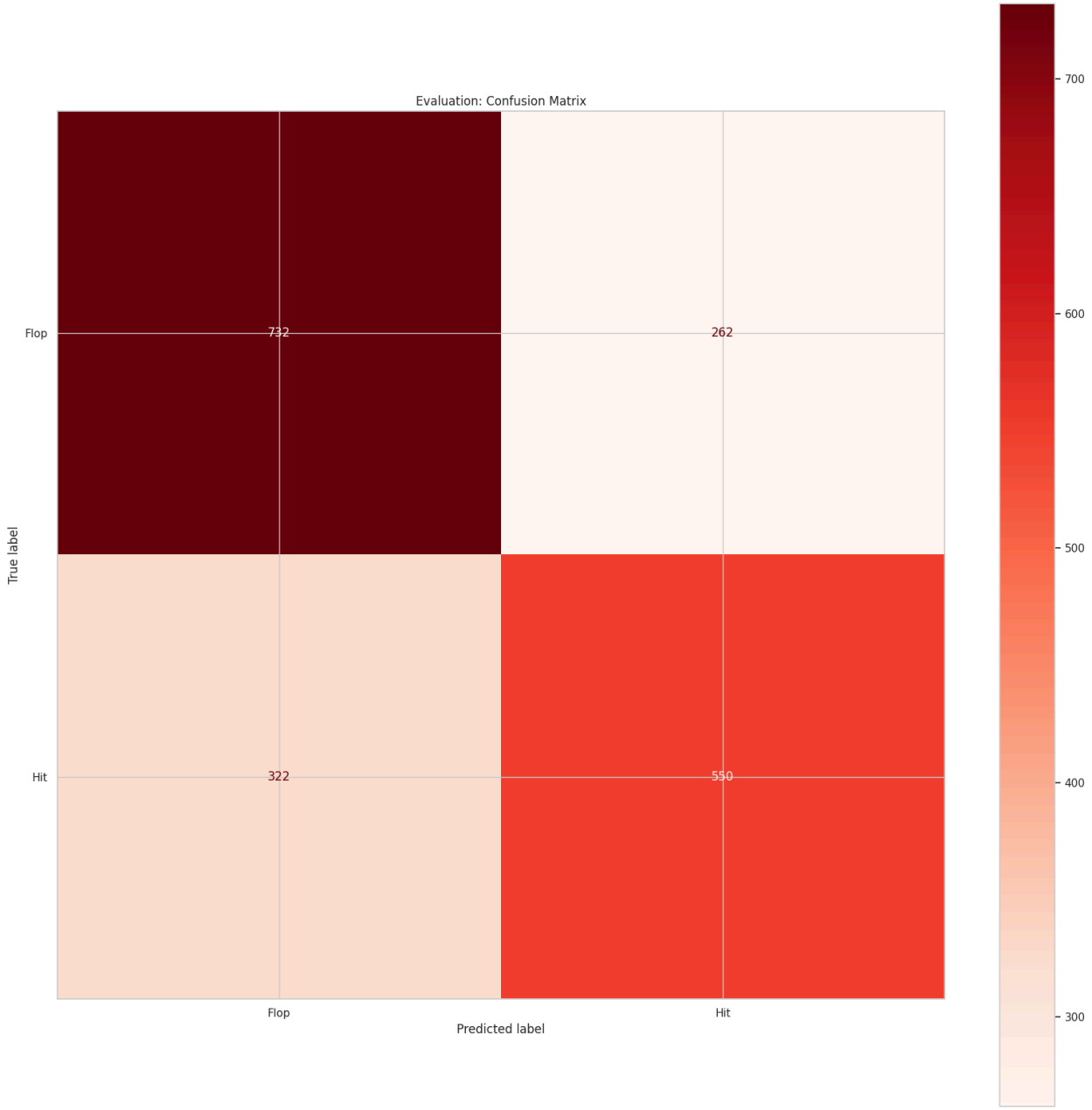


The accuracy score is a far cry from the desired 90% stated in business understanding. The data itself isn't at fault but more so the sore points are the nature of the film industry and the target we are trying to predict which is the exact dollars and cents a film will produce. The average error prediction of $1.10 million is the best I've gotten with all the tweaks I have made in training for the model. We can see the stand out features that the model shows were most important in achieving the prediction.

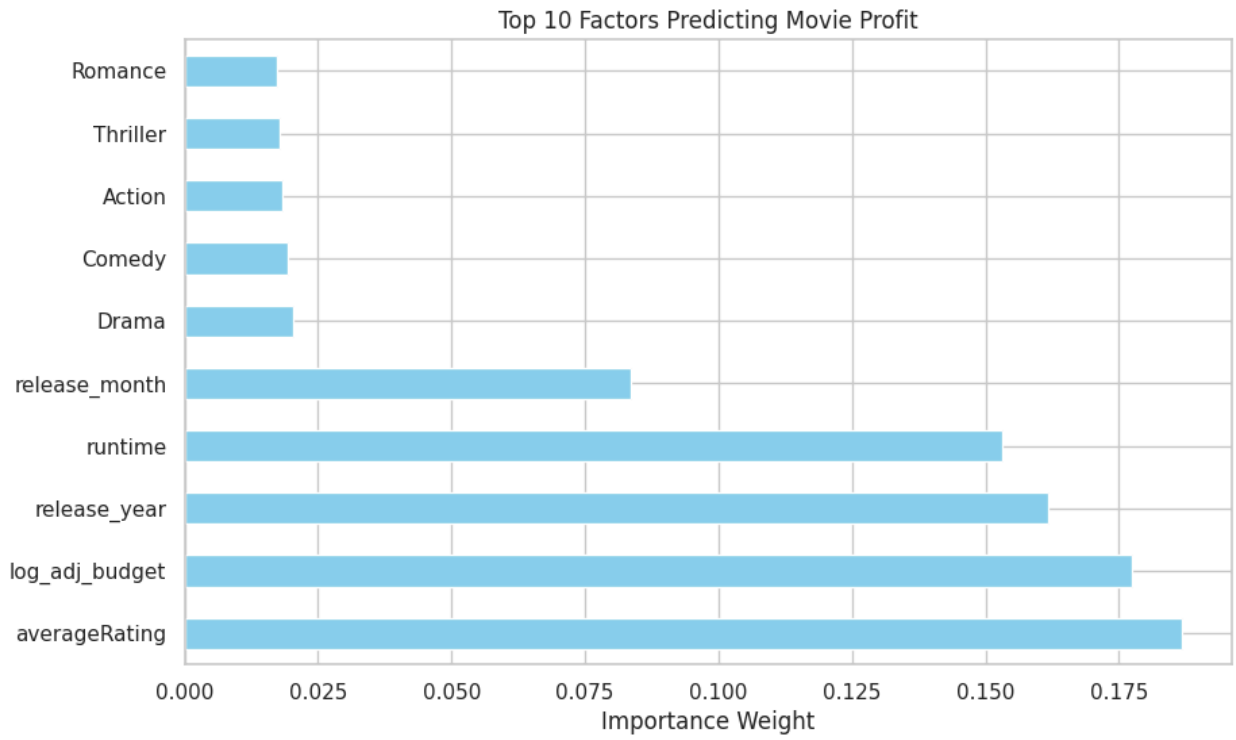
<u>Classification Model Evaluation</u>

Here is where I held my breath for the results given our 73% short fall with the previous model. However, with the results I could breathe normally.

Evaluation Metrics:
Classification Accuracy: 69%
Confusion Matrix: Shows the model accurately identified 732 Movie Flops and 550 Movie Hits. However, it incorrectly classified 584 other films in total: It predicted over-predicted success for 262 flopped films and failed to recognize 322 hit films.



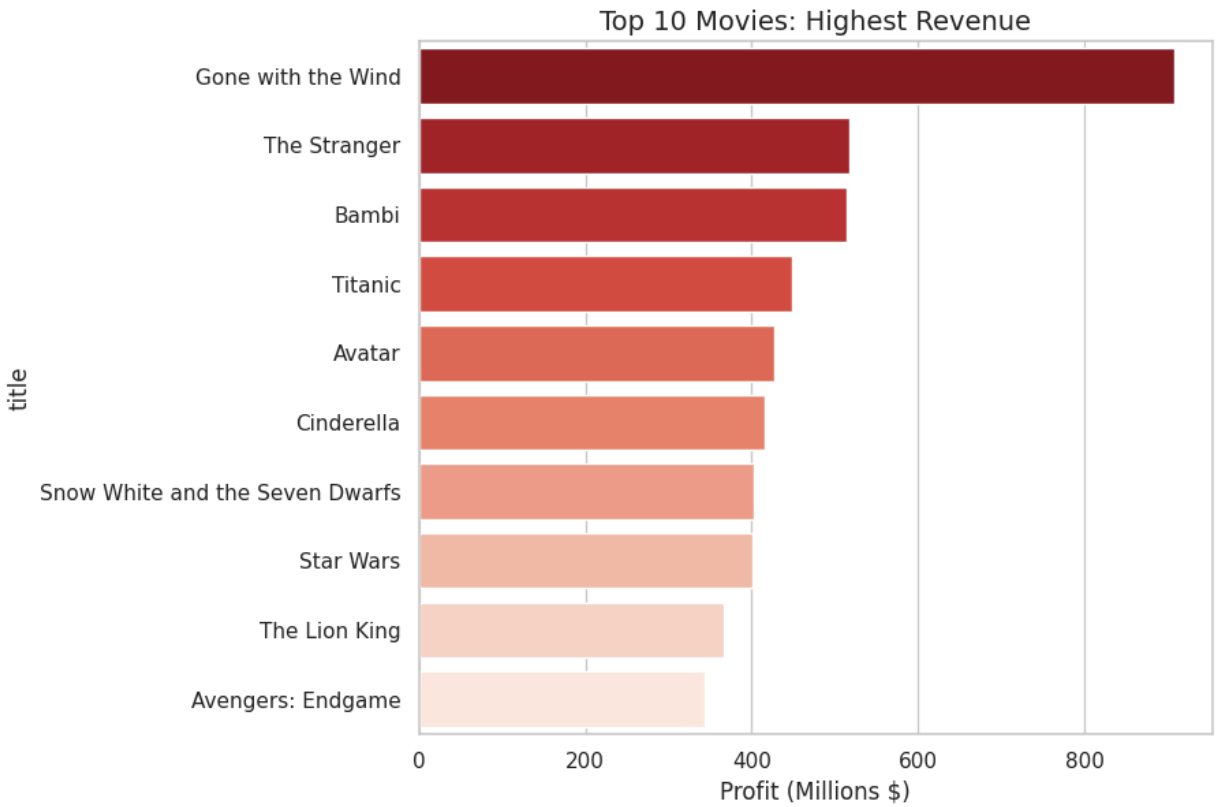
A win is a win lol, but this 69% accuracy tells me that there is much more involved in being able to double the investment in a film.



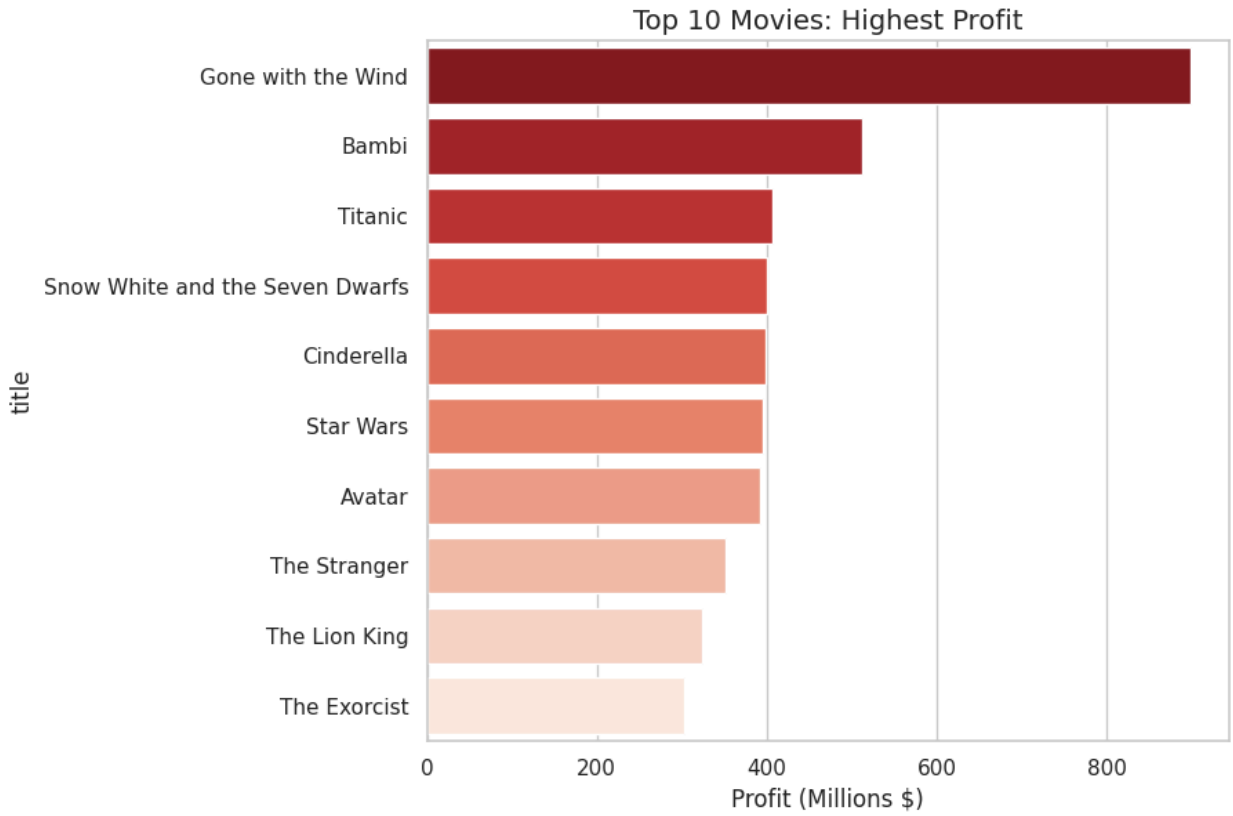
The important features for the prediction are the same as the Regression model's with the exception that average rating out weights the budget.

Deployment (Make the findings or model accessible or production-ready)

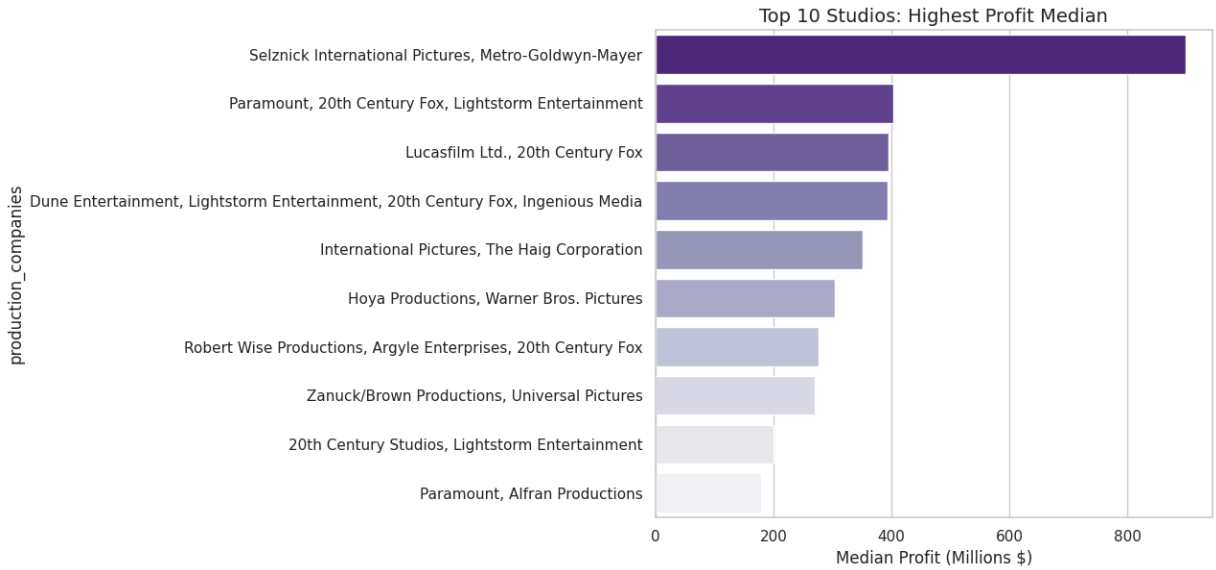
I won't be deploying the model for use given the nature of the non-commercial license use of the data but I will share the dashboard with my findings and my final thought will come in the final stage.



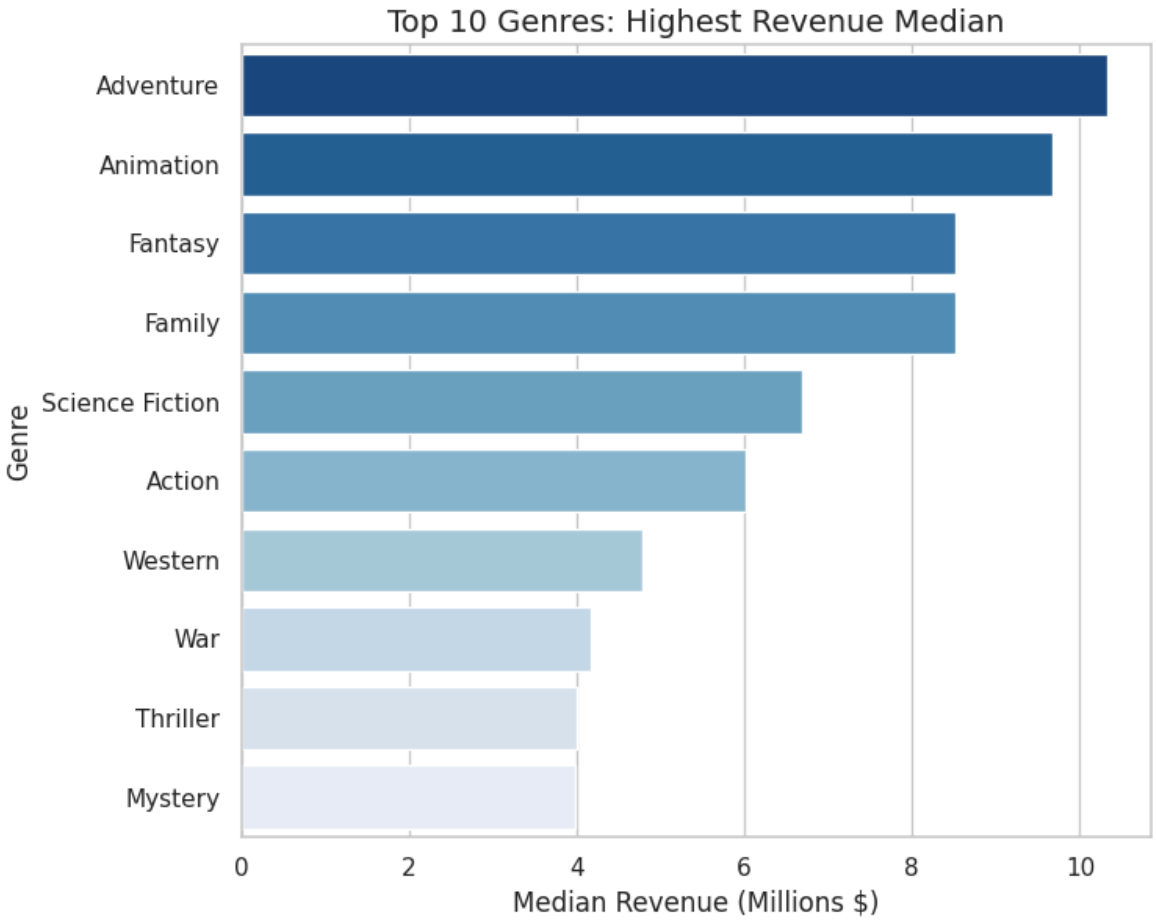
This top 10 list is very different compared to the first list of films I looked at in the Data Understanding stage because we now have an inflation adjusted list.



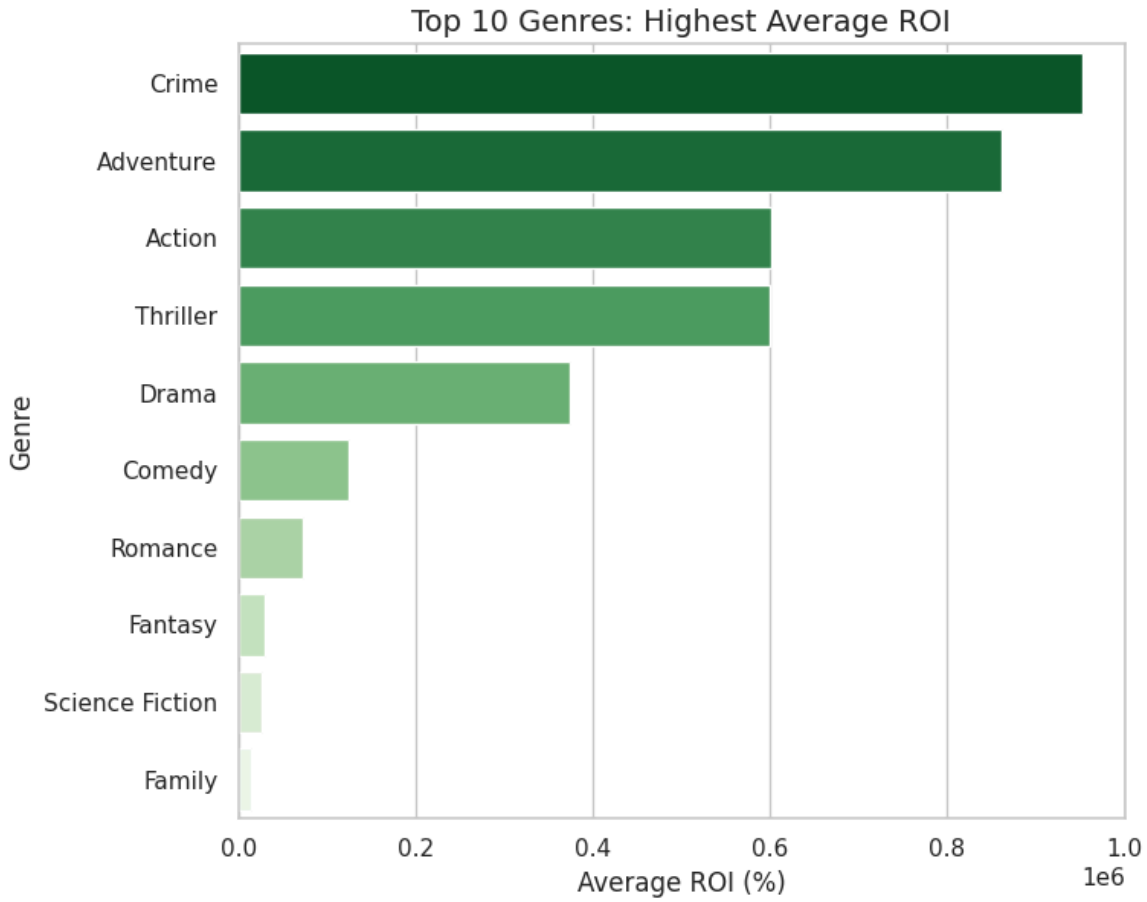
An important thing to look into is which film was considered most profitable given its own budget rather than compare just gross revenue figures of a film.



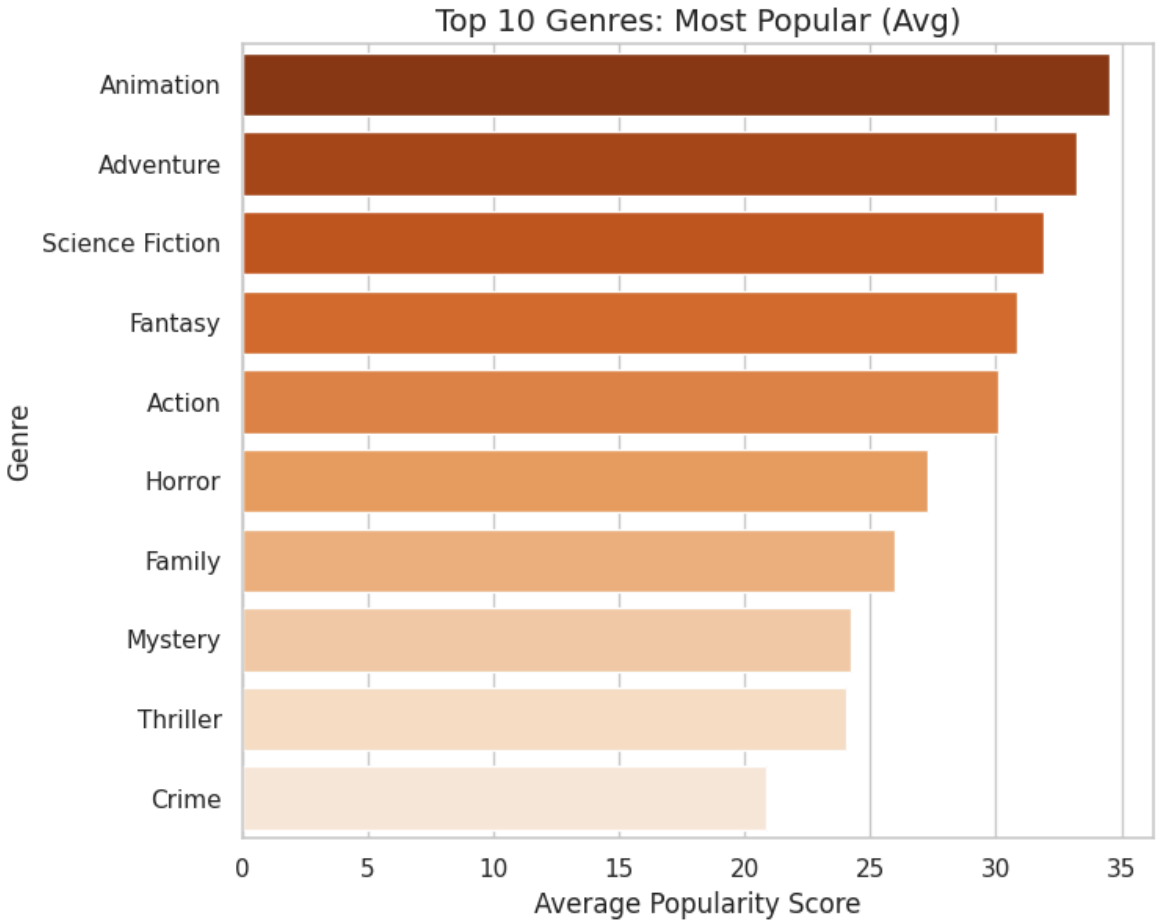
To my surprise my marvel studios is nowhere to be found in the top for films with the highest profit median (average)



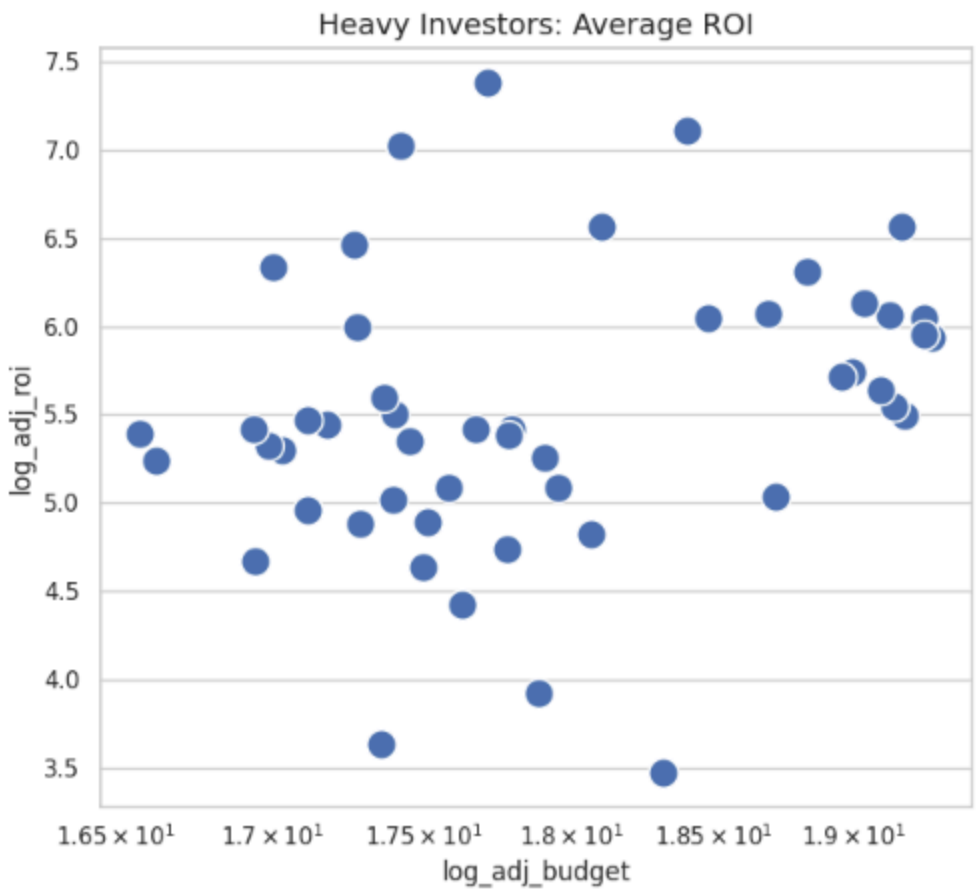
Ladies and gentlemen, the most lucrative movie genres.



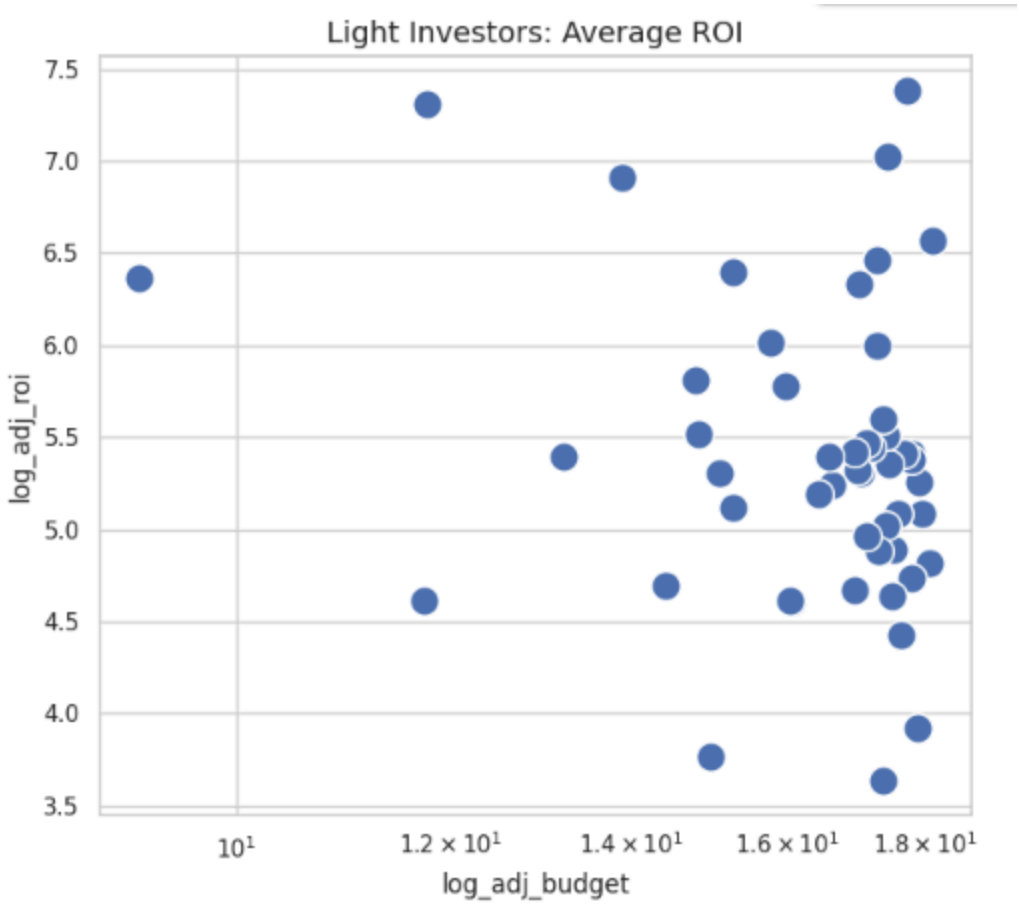
Crime being the top genre for the highest ROI average was a huge surprise to me.



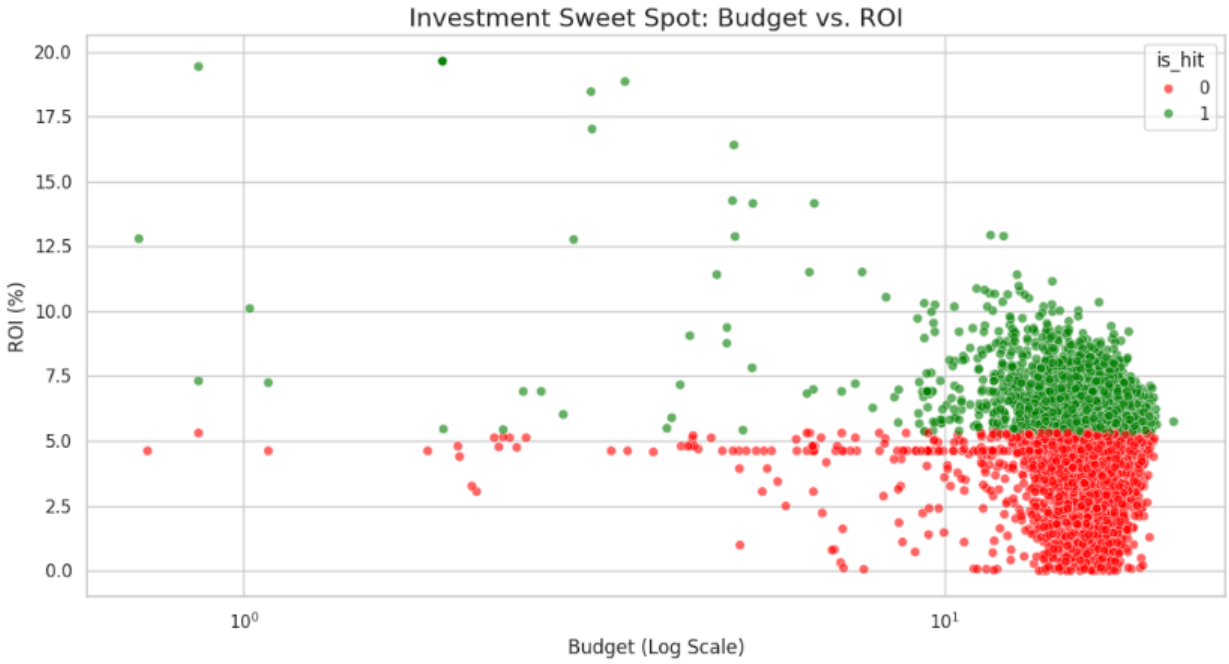
The most popular with the people is Animation, Adventure and Sci-Fi.



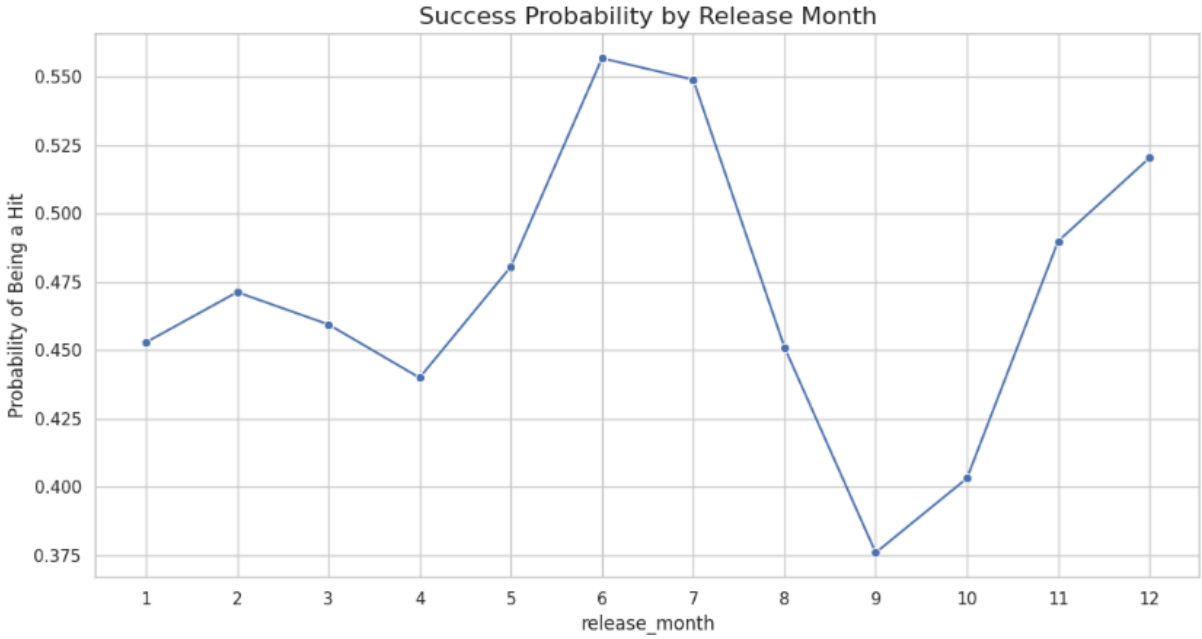
This chart has the top 50 studios that have produced more than 5 films and are considered to be heavy budget investors for their films on average. What is observed is large budget films don’t guarantee an attractive ROI



This chart has the bottom 50 studios that have produced more than 5 films and are considered to be light budget investors for their films on average. What is observed is light budget films don't mean the film will be at a disadvantage to generate an attractive ROI.



Comparing all films' budget and ROI it shows films that have relatively the same budget turns up with a hit or a flop film. We can also see that some of the films with the highest ROI are films that have a lighter budget compared to the large budget films.



June, July, November and December are the release months associated with hit films.

Feedback (Monitor performance and incite continuous development)

Our goal was to achieve 90% accuracy with our predictive model. Our regression model came up way short with an accuracy of 27% and an average prediction error of $1.10 million. It was a difficult task for the model to predict the exact ROI value. The issue isn't the quality of the data nor the quantity though increasing the quantity would improve the accuracy by giving the model more films to learn from.

Our classification model was a better performer and a closer reflection of the film industry's volatile nature. An accuracy of 69% predicting a hit or flop film.

When we take a look at the film's budget vs the ROI I can almost say it's a gamble but I won't because there are more features that can be taken into consideration such as the marketing budget, script & film critic sentiment analysis, the global distribution network and competitor analysis (which other films are sharing the same release period). If I got my hands on this data then we would be looking at a much improved predictive model giving the film executives much more confidence to make decisions on greenlighting a film project.
I found out what makes a film successful from the current dataset but there is much more to it. My classification model got me close to the desired 90% accuracy but that may not be possible given that humans may be habitual but still unpredictable given the unlimited input or influences globally.

As it stands now, what we have is a tool that can set the business foundation for success regarding the film's genre, runtime, release month and budget. The rest is dependent on the creative process and the audience's subjective response to it. This model will give the decision makers the confidence to put all they can in place on the business side giving the creatives the best chance of carrying the film to a success.

This is by setting the film release dates for June, July, November or December, deciding on a runtime suitable to tell the story at hand while setting expectations for the genre being developed and how much effort is needed for chances of creating a hit.

I conclude that while we use math to help us understand the world around us and A.I. using this math to get things done, without human autonomy, curiosity and creativity the tool just sits there. The math has shown us that while some figures and its relationship to others may be constant, the human factor isn't objective. I didn't achieve a 90% accuracy but 69% seems to account for the unquantifiable subjective nature of art that is films and humans the audience.